



Technical Note

Binding mode of the breakthrough inhibitor AZD9291 to epidermal growth factor receptor revealed



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ABSTRACT

The discovery of genetic drivers of lung cancer in patient sub-groups has led to their use as predictive biomarkers and as targets for selective drug therapy. Some of the most important lung cancer drivers are mutations in the EGFR gene, for example, the exon 19 deletions and the L858R variant that confer sensitivity to the front line drugs erlotinib and gefitinib; the acquired T790M variants confer drug resistance and a poor prognosis. A challenge then in targeting EGFR is to produce drugs that inhibit both sensitising variants and resistance variants, leaving wild type protein in healthy cells unaffected. One such agent is AstraZeneca's "breakthrough" AZD9291 molecule that shows a 200-fold selectivity for T790M/L858R over wild type EGFR. Our X-ray crystal structure reveals the binding mode of AZD9291 to the kinase domain of wild type EGFR.

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1. Introduction

Lung cancer is a leading cause of cancer death worldwide and is a highly, and rapidly fatal disease. One of the recent concepts to emerge around lung cancer has been the discovery of genetic drivers in patient sub-groups that can be used clinically as both predictive biomarkers and molecular targets for targeted cancer therapy, for example, EGFR somatic mutations (Cancer Genome Atlas Research Network, 2014; Ding et al., 2008; Huang et al., 2004; Kosaka et al., 2004; Lynch et al., 2004; Paez et al., 2004; Pao et al., 2004). Subsequent to this, a series of randomized clinical trials showed certain patients harboring epidermal growth factor receptor (EGFR) mutations experienced dramatic therapeutic response to EGFR tyrosine kinase inhibitors (Maemondo et al., 2010; Mitsudomi et al., 2010; Mok et al., 2009; Rosell et al., 2012; Zhou et al., 2011). A wide range of different EGFR gene mutations are now associated with the development of lung cancer predominantly in never or former smokers, by activating the EGFR kinase, causing oncogenic transformation (Shigematsu et al., 2005; Yun et al., 2007; Zhang et al., 2006), but also importantly,

conferring sensitivity to tyrosine kinase inhibition. The two most common EGFR mutations are deletions around the LREA motif of exon 19, and the exon 21 L858R point mutation; both instances show sensitivity to the ATP competitive first generation EGFR inhibitors erlotinib and gefitinib (Carey et al., 2006; Huang et al., 2004; Lynch et al., 2004; Paez et al., 2004; Pao et al., 2004). While these drugs show obvious efficacy early in treatment regimens, they are often rendered ineffective by the acquisition of the so-called gate keeper T790M mutation that increases binding affinity for ATP, resulting in drug resistance and a poor prognosis for such patients with the re-occurrence of disease (Pao et al., 2005; Sharma et al., 2007). Development of second generation irreversible inhibitors of EGFR such as canertinib (Allen et al., 2003; Calvo et al., 2004; Smaill et al., 2000), pelitinib (Nunes et al., 2004; Wissner et al., 2003), neratinib (Tsou et al., 2005), dacomitinib (Brzezniak et al., 2013; Engelman et al., 2007; Gonzales et al., 2008) and afatinib (Sanderson, 2013; Solca et al., 2012), offered early promise for the treatment of T790M-mediated acquired resistance (Jänne et al., 2009; Pao et al., 2005). However, preclinical experiments with these compounds in this setting were likely conducted at supra-physiological concentrations that cannot be achieved in human trial due to dose-limiting toxicities (rash and diarrhoea) resulting from inhibition of wild type EGFR in skin and gastrointestinal tissues (Kim et al., 2012; Sos et al., 2010; Zhou et al.,

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2009). There is a clear need for drugs in the clinician's arsenal that will target EGFR kinase harboring the T790M mutation and TKI-sensitizing mutations while leaving the wild type protein uninhibited in non-cancerous cells.

An irreversible EGFR kinase inhibitor, AZD9291, has recently been developed by AstraZeneca (Cross et al., 2014; Finlay et al., 2014) that belongs to a novel class of compounds including WZ-4002 and CO-1686, which are selective for T790M and EGFR TKI-sensitizing mutations over wild type protein. AZD9291 (Fig. 1) shows 200-fold selectivity for T790M/L858R protein over wild type, and has been classed a “breakthrough” compound for fast track development having demonstrated a 64% objective response rate in T790M-positive non-small-cell lung cancer patients that have progressed on a first generation EGFR inhibitor (Janne et al., 2014); it is currently in Phase 3 trials. To understand the molecular basis of this compound's selectivity, we sought to produce crystal structures to show its binding interaction to T790M, T790M/L8587R, and wild type EGFR kinase protein. Our initial efforts have produced a wild type structure by soaking the AZD9291 molecule into preformed crystals of EGFR kinase.

2. Protein production and structure determination

Wild type EGFR protein was synthesized (GenScript) and cloned into the pFastBac HTb plasmid using HindIII and NcoI restriction sites. Baculovirus for wild type protein production were created using the Bac-to-Bac Baculovirus Expression System (Invitrogen, Carlsbad, CA, USA), and were used to infect Sf9 cells. Infected Sf9 cells were harvested 72 h after infection by centrifugation at 600g for 6 min. The cell pellet was resuspended in TBS buffer (25 mM Tris pH 8.0, 137 mM NaCl), snap frozen in liquid nitrogen, and stored at -80°C for later use.

Cell pellets were supplemented with 10 mL of lysis buffer (25 mM Tris.HCl, pH 8.0, 0.5% v/v NP-40 alt, 3 μM RNase A) for every 12–15 g of wet cell mass, and were agitated gently to com-

plete cell lysis. After lysis, the sample was supplemented with glycerol to a 5% final concentration and the NaCl content was adjusted to a final concentration of 150 mM. The lysate was clarified by centrifugation at 20,000g for 30 min in an SS34 rotor, and the supernatant was passed through a 0.45 μm filter (Millipore) before being loaded onto an equilibrated (buffer A: 25 mM Tris.HCl, pH 8.0, 150 mM NaCl, 15 mM imidazole and 5% glycerol) immobilised metal affinity chromatography column (IMAC) (HiTrap IMAC FF, GE Healthcare Life Sciences).

The immobilised protein was washed with buffer A and eluted using a linear gradient of buffer B (25 mM Tris.HCl pH 8.0, 150 mM NaCl, 200 mM imidazole and 5% glycerol). The protein containing fractions were pooled and dialysed overnight at 4°C in dialysis buffer comprising 25 mM Tris pH 8.0, 25 mM NaCl, 1 mM DTT and 5% glycerol, and supplemented with recombinant tobacco etch virus protease for cleavage of the 6His-tag. The protein was subsequently passed over an IMAC column to remove 6His-tag and uncleaved protein. The cleaved protein was collected and loaded on an anion exchange column (HiTrap Q FF, GE Healthcare Life Sciences) pre-equilibrated with QF buffer A (25 mM Tris.HCl pH 8.0, 25 mM NaCl, 2 mM BME, 5% glycerol). The protein was eluted using a linear gradient of QF buffer B (buffer A with 1 M NaCl).

Protein containing fractions from anion exchange were pooled and buffer exchanged into storage buffer (25 mM Tris.HCl, pH 8.0, 150 mM NaCl, and 5% glycerol). Protein was concentrated to approximately 10 mg/mL and was either supplemented to 15% glycerol and stored at -20°C or was used immediately without added glycerol for crystallisation experiments.

EGFR kinase was crystallised using a hanging drop vapour diffusion method by mixing 1 μL protein with 1 μL crystallisation solution comprising 0.1 M MES, 1 M sodium citrate and inverting the drop over 500 μL of the same crystallisation solution. Hexagonal-shaped crystals grew overnight and reached maximum size within one week. An unliganded crystal was flash frozen in liquid nitrogen, cryoprotected with a 70:30 mixture of Paratone N[®] and minerals oils. Data were collected at the Australian synchrotron beamline MX1 using the Blu-Ice software (McPhillips et al., 2002). We were unable to co-crystallise the protein with AZD9291 and so instead resorted to soaking experiments. Another unliganded crystal was soaked overnight in a solution comprising 95% crystallisation buffer supplemented with 5% AZD9291 dissolved in 100% DMSO – the final concentration of AZD9291 in the soak solution was 5 mM. The crystal was cryoprotected in oil, flash frozen, and data were collected on a Rigaku MicroMax 007 HF rotating anode instrument. Data were processed using XDS, POINTLESS, and AIMLESS (Evans and Murshudov, 2013; Kabsch, 2010). Both unliganded and AZD9291 structures were solved by molecular replacement using PHASER and the PDB coordinates 2itx. Structures were refined using REFMAC with ligands and water molecules added to the model using COOT (Emsley et al., 2010; Murshudov et al., 2011). The AZD9291 molecule could be fitted unambiguously into difference electron density in a single orientation and conformation (Fig. 2). The quality of the final AZD9291/EGFR structure at 2.8 Å, and the unliganded structure at 1.9 Å, was assessed using the MOLPROBITY and PDB validation servers (Chen et al., 2010). All data collection and refinement statistics are outlined in Table 1.

3. AZD9291 – reversible binding mode

The crystal structure of the AZD9291/EGFR kinase complex reveals a binding mode similar to the hypothetical model produced by Cross et al. (2014), and with the kinase domain in an active conformation (Fig. 3). The AZD9291 and unliganded structures are very similar overall with an rmsd between 277 overlaid C α atoms

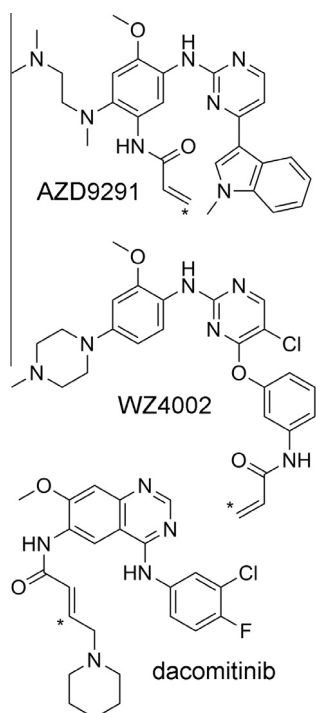


Fig. 1. Molecular structures of AZD9291, WZ4002, and dacomitinib. The sites of alkylation are marked with an asterisk. Figure drawn with ChemDraw (Cambridgesoft).

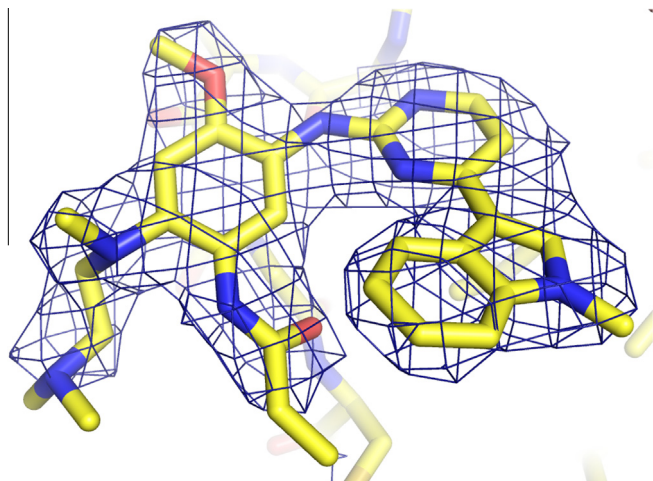


Fig. 2. Electron density map around the modelled AZD9291 molecule. The map is a 2Fo-Fc map contoured at the 1.0 σ level and was calculated before the AZD9291 was included in refinement. Picture drawn using Pymol (Schrödinger).

of 0.36 Å. In our AZD9291 structure, unexpectedly, we do not observe a covalent bond between the acrylamide moiety of AZD9291 and the protein side chain Cys 797 even though the reactive atom centres (AZD9291 C9 and Cys 797 SG) approach to ~3.5 Å. Given the lack of a covalent interaction in our crystal structure and the paucity of information in the literature, we tested the ability of AZD9291 to covalently modify wild type EGFR protein in solution using mass spectrometry. Wild type EGFR

Table 1
X-ray data collection and refinement statistics.

	EGFR	
	AZD9291 complex	Unliganded
<i>Data collection</i>		
Diffraction source	Rotating anode	AS-MX1
Wavelength	1.5418	0.9537
Space group	I23	I23
<i>Unit cell parameters</i>		
$a = b = c$ (Å)	144.89	143.18
Resolution range (Å)	19.4–2.80 (2.95–2.80)	19.9–1.90 (1.94–1.90)
No. unique reflections	12,574 (1809)	38,451 (2574)
Completeness (%)	99.7 (100)	99.9 (100)
Multiplicity	38.1 (38.0)	38.6 (38.0)
R_{pim}	0.041 (0.398)	0.051 (0.702)
Mean I/σ (I)	20.9 (2.3)	15.1 (1.8)
CC1/2	0.999 (0.802)	0.999 (0.801)
Wilson B-factor (Å ²)	53.0	22.4
<i>Refinement</i>		
$R_{\text{work}}/R_{\text{free}}$	0.193/0.244	0.197/0.212
Mean B-factor (Å ²)		
Protein	61.5	30.7
Ligand	69.6 (AZD9291)	55.1 (MES, Na ⁺)
Water	40.4	30.9
<i>No. of atoms</i>		
Protein	2177	2354
Ligand	37	20
Water	4	111
<i>Rmsd</i>		
Bond (Å)	0.012	0.012
Angle (°)	1.58	1.51
<i>Ramachandran</i>		
Favoured (%)	94	99.0
Allowed (%)	6	1.0

Values for the outer shell are given in parentheses.

kinase domain was mixed with a 50% molar excess of AZD9291 and incubated for 3 h in crystallisation buffer before analysis by the University of Auckland, Auckland Science Analytical Services Mass Spectrometry facility by LC-MS (QSTAR XL, Applied Biosystems). The mass spectrometry results (Supporting information Fig. S1) show definitively that wild type protein is covalently modified by AZD9291. What then does our crystal structure inform us about the selectivity of AZD9291 for different variants of EGFR?

In the crystal structure AZD9291 binds on the outer edge of the ATP-binding pocket sandwiched between the P-loop and the protein backbone between Pro 794 and Cys 797. Two hydrogen bonds help to anchor the ligand in place; AZD9291 N4 hydrogen bonds to the hinge atom Met 793 N and the ligand carbonyl oxygen hydrogen bonds to Cys 797 N. The single hydrogen bond to the hinge region is in contrast to the two bonds seen for WZ4002, a compound that shares a similar central core structure to AZD9291 (Zhou et al., 2009).

The substituted phenyl ring of AZD9291 makes a number of van der Waals interactions with the protein backbone below. Extensive interactions with non-polar residues from above complete the sandwiching effect; Leu 718 contacts the AZD9291 phenyl ring, Ala 743 contacts the pyrimidine ring, Val 726 and Phe 723 contact the indole ring. The protein backbone of Leu 718–Gly 719 also forms van der Waals interactions with the edge of the indole ring. A comparison with our own unliganded EGFR structure shows that on AZD9291 binding the only changes induced are in the tip of the P-loop, specifically residues Ser 720, Gly 721, Ala 722, Phe 723, and Gly 724, previously disordered and thus unmodelled, now form a stable and “bent” structure with the Phe 723 side chain located underneath the P-loop, making a 3.7 Å contact with AZD9291. It has been observed that significant selectivity for a particular protein kinase can be achieved through the ability of the P-loop to form a closed or bent structure and also importantly, to contact the ligand through the side chain of the terminal P-loop Phe or Tyr residue – as we see in our structure (Dar et al., 2008; Deininger et al., 2005; Guimarães et al., 2011; Hari et al., 2013; Seeliger et al., 2009; Yosaatmadja et al., 2015). While this observation may explain some degree of selectivity for the EGFR tyrosine kinase it does not explain the pattern of selectivity for EGFR mutants observed in the cell-based and clinical data.

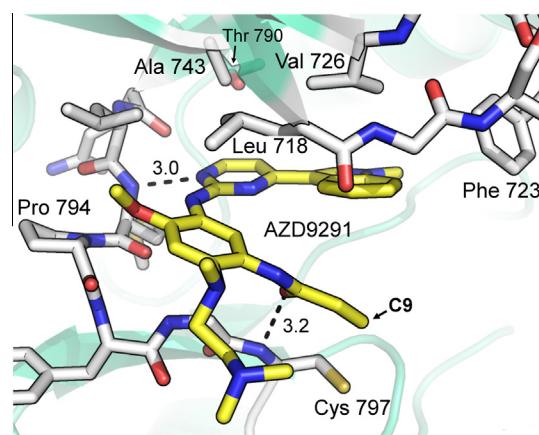


Fig. 3. Binding of AZD9291 (yellow sticks) to EGFR (white sticks and green transparent cartoon structure). Hydrogen bonds are drawn as dashed lines with distances labelled in Å. The side chain of the known covalent attachment site is shown (Cys 797) as well as the reactive centre on the AZD9291 molecule (C9). Selected residues involved in van der Waals interactions with the ligand are highlighted – the AZD9291 molecule is effectively sandwiched on the edge of the ATP-binding site. Figure was drawn with Pymol (Schrödinger).

4. Molecular modelling probing inactive vs active EGFR conformations

Without a crystal structure of AZD9291 bound to the T790M/L858R EGFR kinase domain we have instead modelled this interaction using relevant crystal structures where the P-loop is found in a more closed conformation which would “push” the AZD9291 molecule closer to the reactive cysteine 797. Two structures were used; the complex of T790M EGFR with dacomitinib (PDB ID 4i24) that displays a P-loop conformation most like that within our structure, and also a more relevant variant structure, T790M/L858R in complex with the irreversible inhibitor PD168393 (PDB ID 4llo). In both cases molecular docking (GOLD v1.7, CCDC) produced binding models by including soft potentials for Gly 796 and Cys 797 to allow the close approach of the inhibitor to Cys 797 to mimic covalent attachment.

The modelled T790M (4i24) structure displays an inactive conformation in which the N-terminal and C-terminal lobes of the protein are brought closer together than in the equivalent active conformation (Gajiwala et al., 2013). In our model the P-loop has moved downwards and we have simultaneously maintained, through the inclusion of distance restraints in the modelling protocol, both appropriate van der Waals contact distances from the non-polar side chains to the top of the inhibitor and the AZD9291 N4 hydrogen bond to the hinge region (Fig. 4). The loop movement and conserved interactions in turn “push” AZD9291 towards Cys 797 such that the C9 atom closely approaches the sulfur atom to a distance of ~ 2 Å – modelled as a covalent bond in Fig. 4. Our model also suggests that additional ionic interactions potentially form between AZD9291 atom N2 (presumably protonated at physiological pH) and the protein side chains Asp 800 and Glu 804 that approach the ligand nitrogen to distances of ~ 3.2 Å.

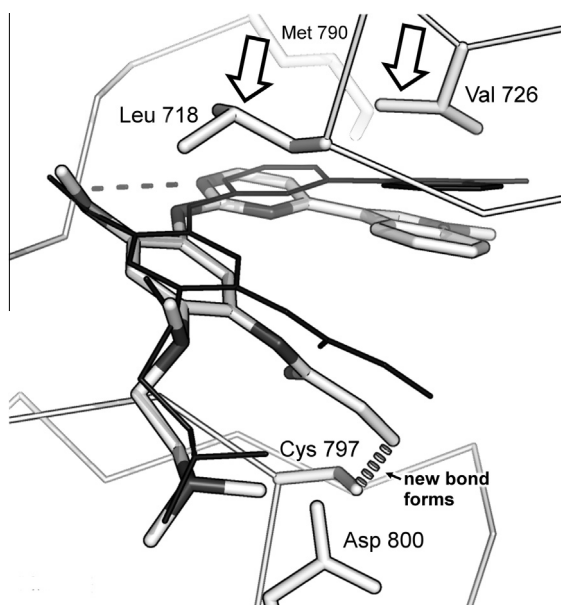


Fig. 4. Hypothetical model of AZD9291 binding to T790M EGFR kinase using the crystal structure of T790M EGFR complexed with irreversible inhibitor dacomitinib as a template (PDB code 4i24). Bold arrows show qualitatively the movement of the P-loop we have modelled. The crystal structure ligand location is drawn in thick black lines while the modelled location of AZD9291 is drawn as a white stick model – a modelled hydrogen bond is shown as a thin, black, dashed line. A covalent bond (thick hashed line) is modelled to Cys 797 and the amine side chain of AZD9291 is shown modelled as approaching Asp 800 and Glu 804 (not shown) where an ionic bond might form. Drawn with Pymol (Schrödinger).

Does AZD9291 binding parallel that of dacomitinib and select for an inactive conformation of the protein? Dacomitinib, like AZD9291, does not form a covalent bond when soaked into wild type crystals with EGFR presumably locked by crystal packing in an active protein conformation. Irreversible dacomitinib behaviour required co-crystallisation with T790M EGFR protein – the crystal structure in this case displaying an unusual inactive conformation (Gajiwala et al., 2013). The authors suggest that dacomitinib selectivity for T790M EGFR and similar variants, is derived from the lower conformational stability of these proteins relative to wild type, basically, their propensity to sample various conformational states including inactive conformations (Gajiwala et al., 2013). Is this the same case for AZD9291 binding in particular for the T790M/L858R variant?

To probe this hypothesis we modelled AZD9291 into the closed P-loop, active conformation of the T790M/L858R complex structure with PD168393 by Brewer et al. (PDB ID 4llo; Brewer et al., 2013). Brewer describes evidence both crystallographic and from patient derived material from lung cancer tissues, that the double mutant protein preferentially conforms to an active state more so than the wild type. Does this suggest that AZD9291 targets an active EGFR kinase domain conformation? Certainly our model derived from the PD168393 structure (produced in the same way as our 4i24-based model) suggests that AZD9291 can bind this active conformation again with a closed P-loop promoting the close approach of the reactive centres of the inhibitor and protein. The most obvious difference in this model compared to our crystal structure is that the tip of the P-loop is extended away from the inhibitor and the terminal Phe residue extends towards solvent space. We contend that the reconfiguration of the P-loop tip that we observe *in crystallo* to a bent conformation with Phe 723 pointing into the active site, is an important and conserved feature of AZD9291 binding that may impact on the exact conformation of the EGFR kinase domain that is favoured whether inactive, active or some other atypical arrangement.

Our crystal structure and molecular modelling suggest that direct contact with either of the mutated amino acids, T790M (Fig. 4) or L858R, has no bearing on the AZD9291 binding mode but instead these mutations only influence protein conformation and molecular dynamics. This contrasts with the proposed selectivity mechanism of WZ4002. WZ4002 is related to AZD9291 by the common central core structure but most obviously differs in having the alkylating functionality on the opposite end of the molecule. For WZ4002, alkylation of EGFR in an active conformation crystal structure is apparent, and selectivity here is proposed through the contact between the inhibitor chloride substituent and the methionine of the T790M protein – we predict no equivalent interaction with the gatekeeper methionine in the AZD9291 system.

We have revealed the reversible binding mode of AZD9291 to wild type EGFR protein and have probed by molecular modelling, hypothetical binding modes to inactive and active conformations of the EGFR kinase domain. Our hypotheses and the questions arising from our structure warrant further investigation and many answers will no doubt be provided by the future determination of crystal structures of T790M and T790M/L858R EGFR complexes with AZD9291.

5. Accession numbers

Coordinates and structure factors have been deposited in the Protein Data Bank with accession numbers 4zau and 4wrg for the AZD9291 complex and unliganded structures, respectively.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jsb.2015.10.018>.

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